



WISCONSIN DEPARTMENT OF NATURAL RESOURCES

2024 Comprehensive Summary Report Little Green Lake, Green Lake County Waterbody Code 162500

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Introduction and Objectives

In 2024, the Wisconsin Department of Natural Resources (WDNR) conducted a comprehensive fish survey of Little Green Lake in order to monitor the fishery and guide fisheries management. The survey included spring fyke netting that targeted northern pike, muskellunge, and walleye along with spring electrofishing to assess bass/panfish. Primary sampling objectives are to characterize species composition, relative abundance, and size structure. This report provides a summary of survey results, including management recommendations.

FYKE NETTING SURVEY INFORMATION

Site Location	Survey Dates	Target Species	Net Nights/Miles Shocked
Little Green (SN1)	March 7 - March 23	Northern pike	34 net nights
Little Green (SN1)	March 7 - April 9	Walleye	115 net nights
Little Green (SE1)	April 9	Walleye	4.5 miles
Little Green (SN2)	March 7 - April 14	Muskellunge	131 net nights
Little Green (SE2)	May 23	Bass/Panfish	4.5 miles bass/1 mile panfish
Little Green (SN2)	2025	Muskellunge	

Metric Descriptions

- Catch per unit effort (CPUE) is an index used to measure fish population relative abundance**, which simply refers to the number of fish captured per unit of distance or time. For netting surveys, we typically quantify CPUE by the number and size of fish per net night. For electrofishing, we quantify CPUE as the number caught per mile of water electrofished. CPUE indexes are compared to statewide data by percentiles and within lake trends. For example, if a CPUE is in the 90th percentile, it is higher than 90% of the other CPUEs in the state.
- Total abundance is a metric that describes population size and is estimated by mark and recapture.** In our study, all captured (northern pike, walleye, muskellunge) were given a partial pectoral fin or caudal clip and released. Each time the nets were checked, all (northern pike, walleye, muskellunge) were examined for a partial pectoral or caudal fin clip. The number of previously captured individuals (i.e., fin clipped) was recorded, and proportions of marked individuals to unmarked individuals were used to estimate the total abundance of the population.
- Proportional Stock Density (PSD) is an index used to describe the size structure of fish populations.** It is calculated by dividing the number of quality size fish by the number of stock size fish for a given species. PSD values between 40 - 60 generally describe a balanced fish population.
- Length frequency distribution (LFD) is a graphical representation of the number or percentage of fish captured by half-inch or one-inch size intervals.** Smaller fish (or younger age classes) may not always be represented in the length frequency due to different habitat usage or sampling gear limitations.
- Mean age at length is an index used to assess fish growth.** Calcified structures (e.g., otoliths, spines or scales) are collected from a specified length bin of interest (e.g., 6.0-6.9 inches for bluegill). Mean age is compared to statewide data by percentile with growth characterized by the following benchmarks: slow (<33rd percentile); moderate (33rd to 66th percentile); and fast (>66th percentile).
- Relative weight is an index used to assess the plumpness (i.e., condition) of fish.** It is calculated by comparing the observed weight of a fish to the standard weight (i.e., predicted average weight) of that fish, given its length. A relative weight of 93 means it has average plumpness/weight compared to other fish of the same length. Relative weights above 93 mean it is more plump than average.

RELATIVE ABUNDANCE — CATCH PER UNIT EFFORT (CPUE) SPRING NETTING I (SNI)

Species	Protocol	Total Captured	CPUE	Units	Statewide Percentile	Lake Class Percentile	Overall Abundance
Northern pike	Spring Netting I	39	1.1	Per net night	37th	50%	Moderate
Walleye	Spring Netting I	986	8.6	Per net night	73rd	80%	High
Muskellunge	Spring Netting II	66	0.5	Per net night	55th	60%	Moderate
Black crappie	Spring Netting 1	12,457	415	Per net night	99th	100%	High
Yellow perch	Spring Netting I	388	12.9	Per net night	76th	85%	High
Bluegill	Spring Netting I	348	11.6	Per net night	62nd	-	Moderate
Black crappie	Spring Electrofishing II	4	4	Per mile	40th	-	Moderate
Bluegill	Spring Electrofishing II	548	548	Per mile	97th	96th	High
Largemouth bass	Spring Electrofishing II	145	32	Per mile	75th	75th	High
Pumpkinseed	Spring Electrofishing II	61	61	Per mile	95th	96th	High
Yellow perch	Spring Electrofishing II	9	9	Per mile	49th	-	Moderate

DNR Contact

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Lake Information

Acre: 466
Max. Depth: 28
Mean Depth: 10
Shoreline Miles: 4.8
Public Access: 3 Boat Landings
Lake Class: Complex Warm Dark

Regulations

Statewide Default all Species
Panfish = 25 total
Walleye = 15" minimum, 3 bag
Bass = 14" minimum, 5 bag
Northern Pike = 26" minimum, 2 bag
Muskellunge = 40" minimum, 1 bag

Survey Method

- The primary objective of the spring fyke netting I survey is to count and measure adult walleye, northern pike, and mark both to estimate abundance. The primary objective of spring electrofishing 1 is another attempt to determine walleye abundance. The primary objective of the spring netting II survey is to count, measure and mark adult muskellunge. The primary objective of the spring electrofishing II survey is to count and measure adult largemouth bass and panfish. Little Green Lake was sampled according to spring netting I (SNI), spring netting II (SNII), spring electrofishing I (SEI) and spring electrofishing II (SEII) protocols as outlined in DNR Fisheries Monitoring Protocols. Other species of fish may be sampled during each survey, but are considered by-catch as part of that survey.
- Fyke nets were deployed in areas of the lake that contained spawning habitat or were likely travel areas of targeted species. All newly captured northern pike, muskellunge and walleye were given a partial fin clip. Population estimates were calculated for walleye and muskellunge. A sample of walleyes were given floy tags while muskellunge were given floy and PIT tags. Aging structures and weights were also taken from a subsample of fish.



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Northern Pike

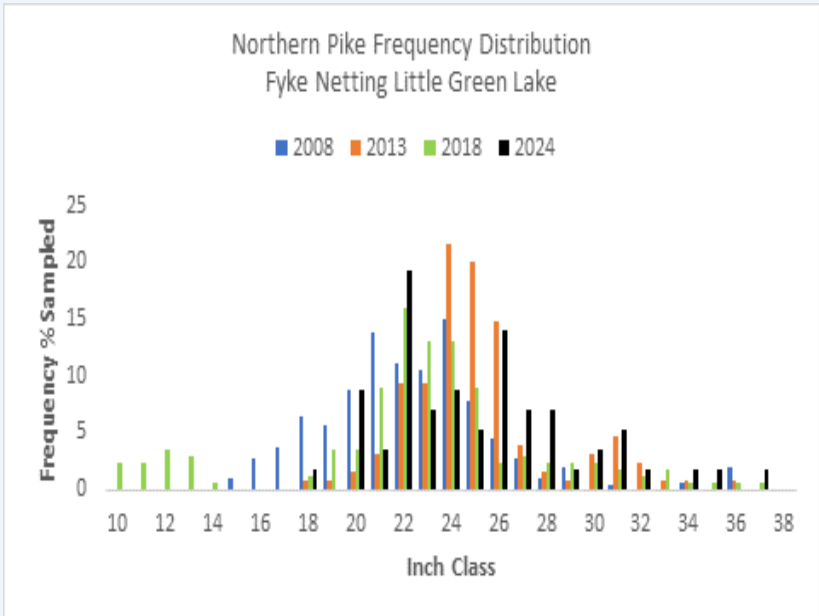
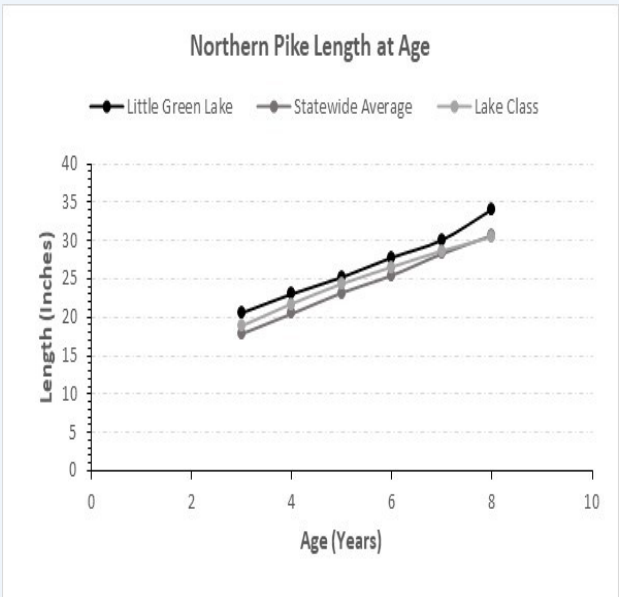
- Northern pike (*Esox lucius*) are a common predatory fish species found across many Wisconsin waterbodies. Northern pike spawn in areas of emergent vegetation at approximately 34-40°F water temperatures. Fyke netting is the preferred sampling gear for northern pike. All results presented for northern pike are from spring fyke netting surveys. Only 1 northern pike was sampled during our SEII survey.

YEAR SIZE STRUCTURE METRICS								
Total Number Measured	Average Length (inches)	Length Range (inches)	Stock and Quality Size (inches)	Stock Number	Quality Number	PSD	Percentile Rank	Size Rating
39	24.7	18.1 –37.2	14.0 and 21.0	37	32	87	93rd	High

SIZE STRUCTURE TRENDS							
PSD by Year				RSD-26 by Year			
2008	2013	2018	2024	2008	2013	2018	2024
72	97	89	87	13	33	22	35

NP Population Estimate (Number Per Acre)				
2008	2013	2018	2024	Historical Average
5.1	2.4	2.9	1.6	3.0

NP Relative Abundance (Number Per Net Night)							
2008	2013	2018	2024	Historical Average	2024 Statewide Percentile	2024 Lake Class Percentile	Rating
5.9	3.6	2.5	1.1	3.3	37th	50%	Moderate





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Northern Pike Species Summary

- Catch rates for northern pike were down from past surveys. Little Green Lake ranks in the 37th percentile compared statewide and in the 50th percentile compared to lakes in the same lake class.
- The population estimate of 746 adult northern pike or 1.6 per acre reflects the lower catch rate in our nets. The population of northern pike is down from previous recent surveys but this variability is no uncommon.
- Average length for northern pike was 24.7 inches with males being 22.4 inches and females 28.1 inches. The PSD value of 87% ranks in the 93rd percentile statewide and has been this high or higher since the population decreased after the 2008 survey. A $RSD-26 = 35\%$ means that 35 percent of the fish caught over 14 inches were also of legal size or larger. This percentage has also increased since the 2008 survey. The largest northern pike captured during our survey was 37.2 inches.
- Growth rates for northern pike were above average when compared to lakes statewide and of the same class.

Walleye Species Summary

- We sampled a total of 723 walleye during our netting survey of Little Green Lake. Total catch rates for walleye were down from the 2018 survey but relatively unchanged from the prior surveys in 2003, 2008 and 2013. The 2024 catch rates ranked in the 73rd percentile compared to statewide rates and in the 80th percentile when compared to lakes in the same lake class (Complex Warm Dark).
- The population estimate of 2.3 per acre attained in 2024 was down some from the past three surveys and down slightly from the 3.1 per acre recent average.
- Average length of all walleyes caught in our netting survey was 15.9 inches. Of the sexually mature fish the average length for males was 18.4 inches and females was 24.0 inches. The PSD value of 55 is down from past surveys and ranks in the 37th percentile statewide. The large number of 10-12 inch fish sampled during our netting survey significantly impacted the PSD value. This just means there are a large number of fish that will be reaching the legal harvestable size in 2025 and 2026. The $RSD20$ value of 38% is near the recent average of 41% and up slightly from the last survey in 2018; it has fluctuated up and down throughout the past 5 surveys.
- Growth rates for walleye were above average when compared to lakes of the same class and when compared to growth rates statewide. Fish reach the legal size of 15 inches during their 4th year. The oldest fish we sampled was a 23.0 inch male that was at least 15 years old and a 27.2 inch female that was at least 12 years old.
- We floy tagged 188 walleye on Little Green Lake during our surveys in 2024. Tag returns resulted in an exploitation rate of 13.4% in the first year. To account for tag loss we used the Lake Winnebago walleye reward study tag reporting correction (33.2%) and a tag loss of 4.7%. The walleye bag limit statewide including Little Green Lake changed April 1, 2024 from 5 fish 15 inches and larger to a 3 bag and 15 inches.

Muskellunge Species Summary

- Catch rates for muskellunge on Little Green Lake have been relatively steady since the survey in 2003 ranging from 0.4 to 0.6 per net night. Catch rates in 2024 were right at the recent average of 0.5 per net night. Little Green ranks in the 55th percentile compared statewide and in the 60th percentile compared to lakes in the same lake class.
- The population estimate of 186 adult musky or 0.4 per acre is relatively high but down from survey in 2018. These assessment criteria place Little Green Lake as a musky class B lake.
- Average length for muskellunge was 39.9 inches with males being 38.0 inches and females 41.9 inches. The PSD value of 100% ranks in the top percentile statewide and has been 100% since our 2003 survey. The average size has increased for muskies over all, but females are still down from the 46.4 inch average in 2003 though this average was taken from a relatively small sample of fish. $RSD-40 = 42\%$ means that 42 percent of the fish caught over 20 inches were also of legal size or larger. This percentage has increased yearly since the 2003 survey. No fish 50 inches or larger have been sampled since the 2003 survey. The largest musky sampled during our survey was a 46.0 inch female.
- Little Green Lake was one of the lakes studied from 2013 to 2016 as a part of the pair feeding study. The research compared post-stocking survival for musky raised on pellet food with survival of those raised on minnows. Little Green Lake had fish stocked in 2013, 2014 and 2016; a total of 932 pellet raised fish and 466 minnow raised. In 2024, we caught 17 pellet raised and 6 minnow raised fish, indicating both feeding methods contributed to the population. Statewide, across study lakes the results showed greater success for minnow raised fish. The study also concluded that if musky gametes are collected from southern Wisconsin the fish get an earlier start and can begin feeding on pellets and later convert to minnows. This method can save costs for food and produce a fall fingerling musky similar in size to fish raised only on minnows.
- Growth rates for muskellunge on Little Green Lake were average to slightly below average when compared to lakes regionally and of the same lake class. On average it takes 10 years for musky to reach the legal size of 40 inches with females reaching it in 8 years and males reaching it in 11.



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Walleye

- Fyke netting is the preferred sampling gear for walleye when water temperatures reach 45F. Relative abundance is reported by sampling location and year below. All size structure and population estimate results presented below are from walleye fyke netting. Similar sampling methods were used in previous surveys that allows for comparison between years.

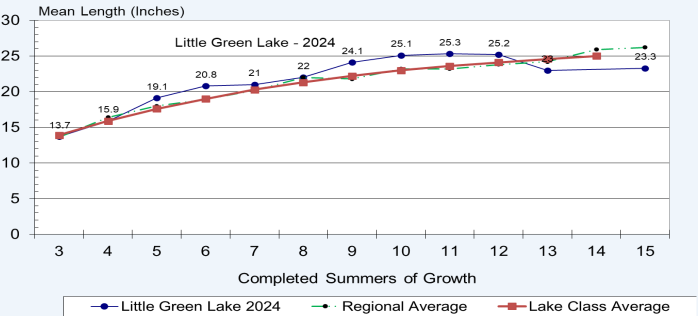
YEAR SIZE STRUCTURE METRICS								
Total Number Measured	Average Length (inches)	Length Range (inches)	Stock and Quality Size (inches)	Stock Number	Quality Number	PSD	Percentile Rank	Size Rating
723	19.3	6.9–28.4	10.0 and 15.0	603	331	55	37th	Low

SIZE STRUCTURE TRENDS											
PSD by Year						RSD-20 by Year					
2003	2008	2013	2018	2024	Historical Average	2003	2008	2013	2018	2024	Historical Average
95	82	87	79	55	79.6	55	29	51	32	38	41

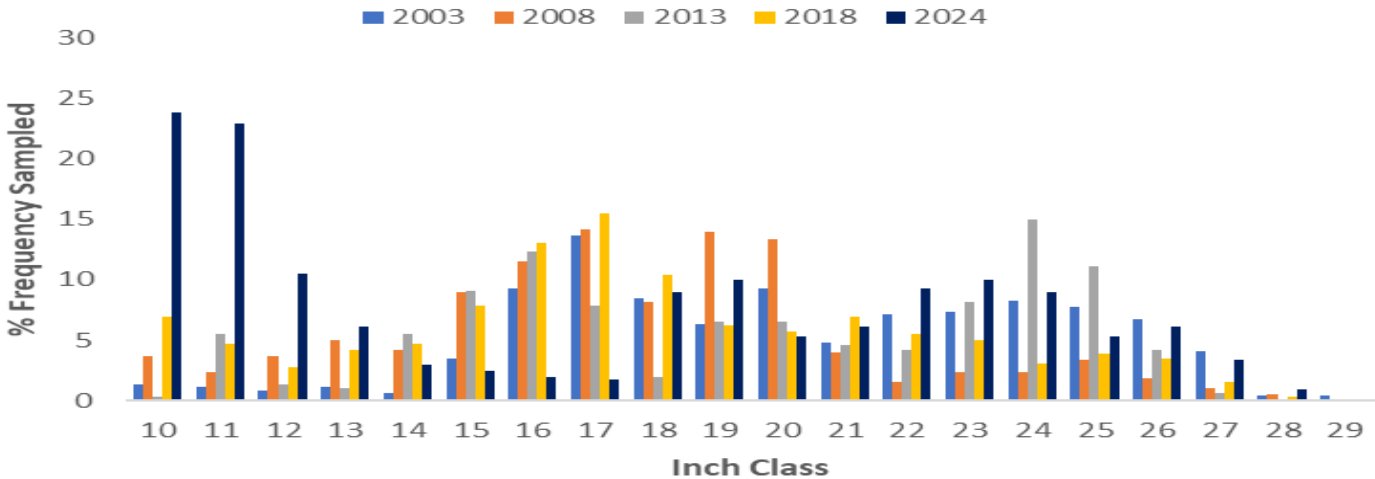
Walleye Population Estimate (Number Per Acre)				
2008	2013	2018	2024	Historical Average
3.6	3.2	3.3	2.3	3.1

Total Walleye Relative Abundance (Number Per Net Night)								
2003	2008	2013	2018	2024	Historical Average	2024 Statewide Percentile Rank	2024 Lake Class Percentile Range	2024 Abundance Rating
11	10	10	21	9	12.2	73rd	80th	High

Walleye Growth Rates
Little Green Lake, Green Lake Co.



Walleye Frequency Distribution - SN1 Little Green Lake





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Muskellunge

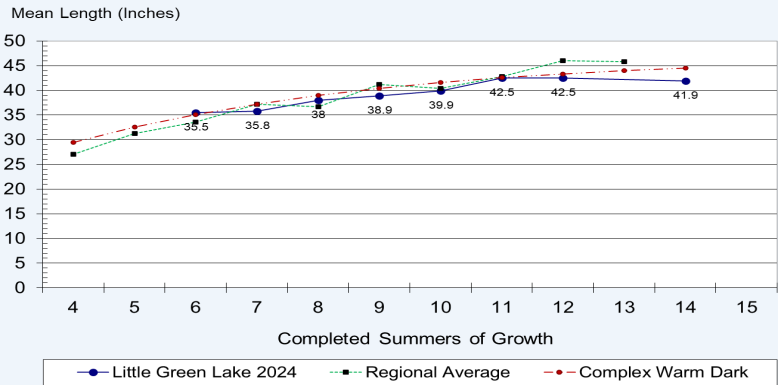
- Fyke netting is the preferred sampling gear for muskellunge when water temperatures reach 50, however, in some waterbodies spawning may begin at ice out. Relative abundance is reported by sampling location and year below. Size structure presented below are from overall muskellunge fyke netting efforts. Similar sampling methods were used over the years to allow for comparison.

YEAR SIZE STRUCTURE METRICS SPRING NETTING																	
Total Number Measured	Average Length (inches)		Length Range (inches)		Stock and Quality Size (inches)		Stock Number	Quality Number	PSD	Percentile Rank	Size Rating						
66	39.9		34.2 –46.0		20.0 and 30 .0		66	66	100	98th	High						

SIZE STRUCTURE TRENDS																	
PSD30 by Year					RSD-40 by Year					RSD-45 by Year					RSD-50 by Year		
2003	2008	2013	2018	2024	2003	2008	2013	2018	2024	2003	2008	2013	2018	2024	2003	Other Years	
100	100	100	100	100	41	19	25	26	42	22	5	0	5	7	3	0	

Muskellunge Population Estimate (Number Per Acre)						Muskellunge Relative Abundance (Number Per Net Night)										
2018	2024	Historical Average	2024 Statewide Percentile Rank	2024 Lake Class Percentile Range	2024 Abundance Rating	2003	2008	2013	2018	2024	Historical Average	2024 Statewide Percentile Rank	2024 Lake Class Percentile Range	2024 Abundance Rating		
0.86	0.4	0.63	75	NA	75	0.4	0.4	0.6	0.6	0.5	0.5	55th	60th	Moderate		

Muskellunge Growth Rates
Little Green Lake, Green Lake Co.



Muskellunge Average Length Inches					
Sex	2003	2008	2013	2018	2024
All	37.4	37.2	37.8	38.9	39.9
Male	38.1	37	35.9	37	38
Female	46.4	38.3	40.9	40.5	41.9

Muskellunge Length Distribution - Fyke Netting Little Green Lake

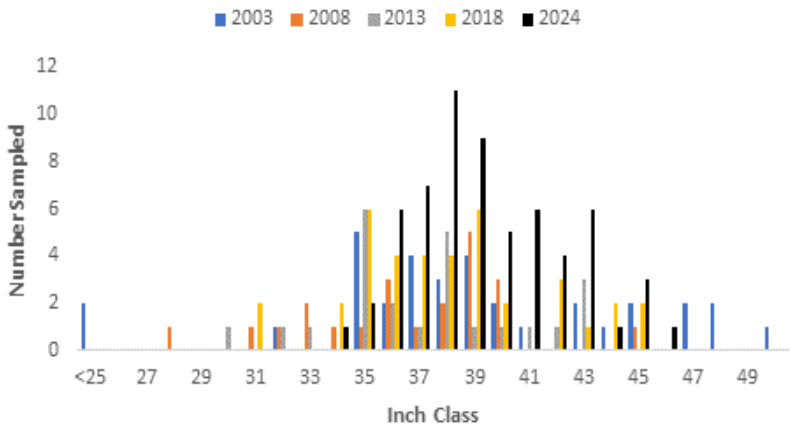


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Largemouth Bass

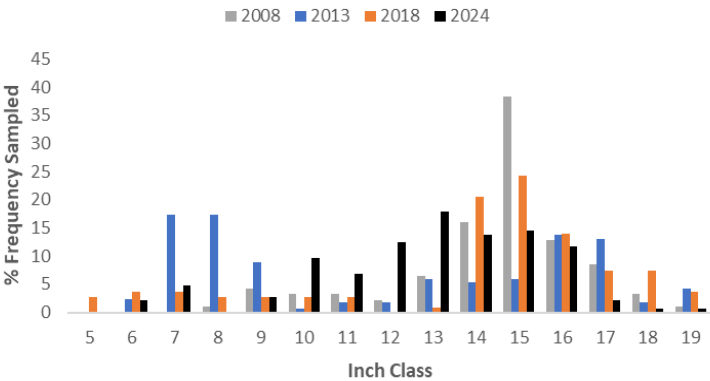
- Largemouth bass (*Micropterus Salmoides*) are a common predatory fish species found in many Wisconsin waterbodies. Largemouth Bass typically spawn in shallow nearshore areas consisting of sand/mud or gravel substrate at approximately 60-70°F water temperatures. Electrofishing is the preferred sampling gear for bass. All results presented for largemouth bass are from spring electrofishing surveys.

YEAR SIZE STRUCTURE METRICS								
Total Number Measured	Average Length (inches)	Length Range (inches)	Stock and Quality Size (inches)	Stock Number	Quality Number	PSD	Percentile Rank	Size Rating
145	13.3	6.5 –19.4	8.0 and 12.0 inches	135	107	79	80th	High

RELATIVE ABUNDANCE (CPUE = NUMBER PER MILE) TRENDS							
2008	2013	2018	2024	Historical Average	2024 Statewide Percentile Rank	2024 Lake Class Percentile Range	2024 Abundance Rating
23.5	36.8	22.3	32.2	28.7	75th	75th	High

SIZE STRUCTURE (PSD) TRENDS						SIZE STRUCTURE (RSD 14 INCH) TRENDS				
PSD by Year				Historical Average	Historical Percentile	PSD by Year				Historical Average
2008	2013	2018	2024			2008	2013	2018	2024	
88	64	88	79	80	81st	80	55	86	47	67

Largemouth Bass Length Distribution - SE2 Little Green Lake



AVERAGE AGE AT SPECIFIC LENGTH 14 Inch Bin							
Specie	Sex	Count	Average Length	Average Age	Age Range	Regional Rating	Lake Class
Largemouth Bass	Male	4	14.4	5.8	4 - 9	Above Average	Average
Largemouth Bass	Female	6	14.4	5.8	4 - 8	Above Average	Average
Largemouth Bass	All	11	14.3	5.8	4 - 9	Above Average	Average

Species Summary

- Little Green Lake supports a moderately high density largemouth bass population with catch rates of 32.2 fish per mile of electrofishing. A catch rate of 32.2 fish per mile ranks in the 75th percentile among lakes throughout Wisconsin and in the 75th percentile when comparing to lakes from the same lake class (Complex Warm Dark). Relative abundance comparisons among years indicate that abundance is slightly above the historical average.
- Size structure of largemouth bass was high with a PSD value of 79% which ranks in the 80th percentile when compared to statewide values. When compared to recent surveys largemouth bass PSD values are down slightly from most recent survey in 2018 but right at the historical average.
- Growth metrics calculated from age estimates indicate that largemouth bass have above average growth rates when compared to regional rates, and average growth rates when compared to lakes in the same lake class. Largemouth bass reach legal size of 14 inches in about over 5 years.



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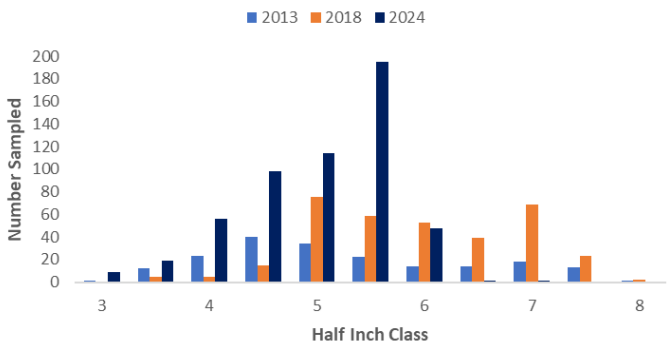
Bluegill

- Bluegill (*Lepomis macrochirus*) are a popular and very common panfish species distributed widely across many Wisconsin waterbodies. Bluegill typically spawn in nearshore areas consisting of sand/mud or gravel substrate at approximately 67-80°F water temperatures. Electrofishing and fyke netting can be effective gear for sampling bluegill, but electrofishing is the protocol and the data we used in this report.

YEAR SIZE STRUCTURE METRICS

Gear	Total Number Measured	Average Length (inches)	Length Range (inches)	Stock and Quality Size (inches)	Stock Number	Quality Number	PSD	Percentile Rank	Size Rating
Electrofishing	548	5.2	1.7 –7.0	3.0 and 6.0	541	50	9	12th	Low

Bluegill Length Distribution - SE2 Little Green Lake



SIZE STRUCTURE (PSD) TRENDS

PSD by Year			Historical Average	Historical Percentile
2013	2018	2024		
31	54	9	31.3	47th

RELATIVE ABUNDANCE (CPUE = NUMBER PER MILE) TRENDS
Fish ≥ 3 inches

2013	2018	2024	Historical Average	2024 Statewide Percentile Rank	2024 Lake Class Percentile Range	2024 Abundance Rating
192	345	541	359	97th	96th	High

AVERAGE AGE AT SPECIFIC LENGTH 6 Inch Bin

Specie	Sex	Count	Average Length	Average Age	Age Range	Regional Rating	Lake Class
Bluegill	Male	10	6.2	4.1	4-5	Above Average	Above Average
Bluegill	Female	12	6.7	6.5	5-8	Below Average	Below Average
Bluegill	All	22	6.5	5.4	4-8	Average	Average

Species Summary

- Little Green Lake in 2024 supports a high density bluegill population with catch rates of 541 fish per mile of electrofishing from the boom shocking survey. Catch rates of 541 per mile is high when compared to all lakes statewide along with lakes in the same lake class.
- Size structure of bluegill was characterized as low based on data from the electrofishing survey. Length data collected from the electrofishing survey resulted in a PSD value of 9 which is in the 12th percentile when compared to statewide values.
- Populations appear to be increasing over recent years. There were a total of 192 bluegills sampled in the 2013 survey and 345 bluegills sampled in 2018 compared to 548 in 2024.
- Growth metrics calculated from age estimates indicate that bluegill have average growth rates when compared to growth rates from bluegill populations across the region in general and from the same lake class (complex warm dark). Growth metrics were below average for females and above average for males.
- We tagged all captured bluegills 6 inches and larger over the survey but with the low size structure there were only 91 fish given a floy tag during the spring survey. Two tag returns reported over the year and neither angler reported harvesting the fish. Due to low returns and low initial sample size we could not estimate annual exploitation.



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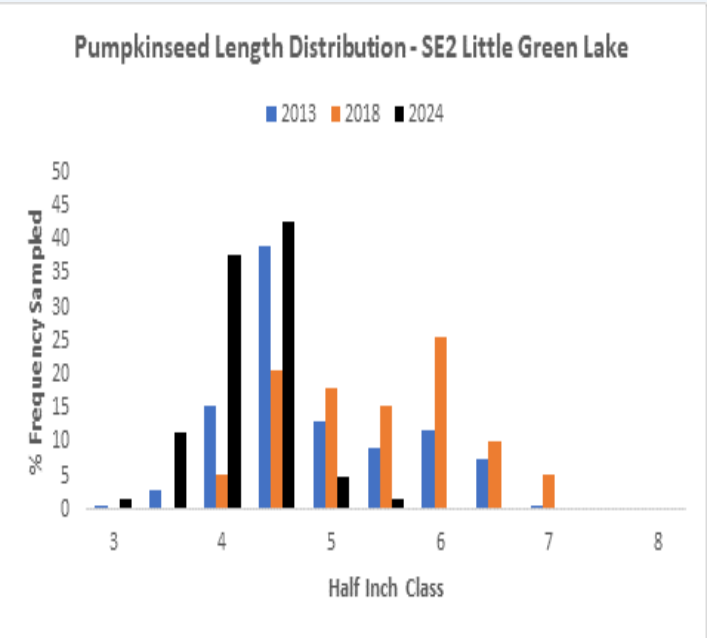


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Pumpkinseed

- Pumpkinseed (*Lepomis gibbosus*) are a common panfish species distributed widely across many Wisconsin waterbodies. Pumpkinseed typically spawn in nearshore areas consisting of sand or gravel substrate at approximately 60-70°F water temperatures. Electrofishing and fyke netting can be effective gear for sampling pumpkinseeds, but electrofishing is the protocol and the data we used in this report.

YEAR SIZE STRUCTURE METRICS									
Gear	Total Number Measured	Average Length (inches)	Length Range (inches)	Stock and Quality Size (inches)	Stock Number	Quality Number	PSD	Percentile Rank	Size Rating
Electrofishing	61	4.4	3.2 - 5.9	3.0 and 6.0	61	0	0	3rd	Low



SIZE STRUCTURE (PSD) TRENDS				
PSD by Year			Historical Average	Historical Percentile
2013	2018	2024		
20	41	0	20.3	35th

RELATIVE ABUNDANCE (CPUE = NUMBER PER MILE) TRENDS						
2013	2018	2024	Historical Average	2024 Statewide Percentile Rank	2024 Lake Class Percentile Range	2024 Abundance Rating
144	39	61	81.3	95th	96th	High



Photo Wisconsin DNR

Species Summary

- In 2024 the lake supports a high density pumpkinseed population with catch rates of 61 fish per mile of electrofishing from the boom shocking survey. Catch rates of 61 per mile is high when compared to all lakes statewide along with lakes in the same lake class.
- Size structure was characterized as low based on data from the electrofishing survey. Length data collected from the electrofishing survey resulted in a PSD value of 0 which is in the 1st-6th percentile when compared to statewide values.
- Populations appear to be fluctuating over the recent years. There were a total of 144 pumpkinseeds sampled in the 2013 survey and 39 pumpkinseeds sampled in 2016 compared to 61 in 2024.



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Black Crappie

Black crappie (*Pomoxis nigromaculatus*) are a common panfish species distributed widely across many Wisconsin waterbodies. Black crappie typically spawn in nearshore areas consisting of detritus, sand/mud or gravel substrate at approximately 58-68°F water temperatures. Electrofishing and fyke netting can be effective sampling gear for black crappie and therefore, some results from both gears are presented.

YEAR SIZE STRUCTURE METRICS

Gear	Total Number	Average Length	Length Range	Stock and Quality	Stock Number	Quality	PSD	Percentile	Size Rating
Fyke Netting	315	5.5	4.2 -9.8	5.0 and 8.0	195	25	13	8th	Low
Electrofishing	4	7.9	7.0 -9.0	5.0 and 8.0	Too	Few	Fish	-	-

SIZE STRUCTURE (PSD) TRENDS NETTING

PSD by Year			Historical Average	Historical Percentile
2013	2018	2024		
96	10	13	39.7	32nd

RELATIVE ABUNDANCE (CPUE = NUMBER PER NET NIGHT) TRENDS

2008	2013	2018	2024	Historical Average	2024 Statewide Percentile Rank	2024 Lake Class Percentile Range	2024 Abundance Rating
2.7	9.1	224	415	162.7	99th	100th	High

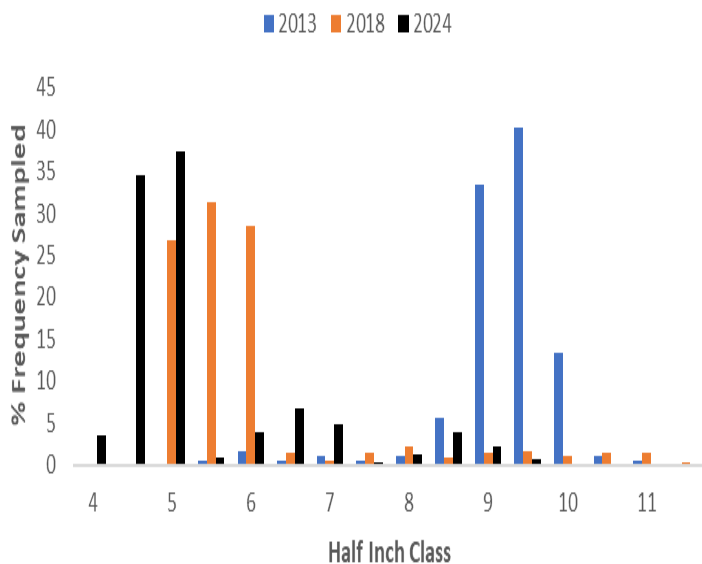
AVERAGE AGE AT SPECIFIC LENGTH 8 INCH BIN

Species	Sex	Count	Average Length	Average Age	Age Range	Regional Rating	Lake Class
Black Crappie	Male	3	8.4	7.3	6 - 8	Below Average	Below Average
Black Crappie	Female	10	8.6	8.1	8 - 9	Below Average	Below Average
Black Crappie	All	13	8.6	7.9	6 - 9	Below Average	Below Average



Photo Wisconsin DNR

Black Crappie Length Distribution - SN1 Little Green Lake



Species Summary

- Black crappie are known for their boom and bust populations and the surveys we use to assess our fisheries rarely give a reliable idea of the existing crappie population. Numbers presented are being used to compare to our previous surveys.
- Little Green Lake in 2024 supports a high density black crappie population with catch rate of 415 fish per net night. Catch rates of 415 fish per net night is high when compared to lakes statewide and same lake class.
- Size structure of black crappie was low based on data from the netting survey. Length data collected from the netting survey resulted in a PSD value of 13 which is in the 8th percentile when compared to statewide values.
- Population trends have been upward over the last two surveys on Little Green Lake. Black crappie populations are well known for fluctuating greatly from year to year but have grown significantly from 9.1 per net night in 2013 to 415 per net night in 2024.
- Growth metrics calculated from age estimates indicate black crappie in have below average growth rates when compared to black crappie across the state and by lake class.



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Yellow Perch

Yellow perch (*Perca flavescens*) are a common and very popular panfish species distributed widely across many Wisconsin waterbodies. Yellow perch typically spawn in nearshore areas consisting of wood and plants shortly after ice out at water temperatures of 45-52°F. Electrofishing and fyke netting can be effective sampling gear for yellow perch and therefore, some results from both gears are presented.

YEAR SIZE STRUCTURE METRICS

Gear	Total Number	Average Length	Length Range	Stock and Quality	Stock Number	Quality	PSD	Percentile	Size Rating
Fyke Netting	237	6.4	5.1 - 10.9	5.0 and 8.0	237	16	7	33rd	Moderately Low
Electrofishing	9	6.4	5.0 - 9.3	5.0 and 8.0	Too	Few	Fish	-	-

SIZE STRUCTURE (PSD) TRENDS NETTING

PSD by Year		Historical Average	Historical Percentile
2013	2024		
8	7	7.5	

RELATIVE ABUNDANCE (CPUE = NUMBER PER NET NIGHT) TRENDS

2008	2013	2018	2024	Historical Average	2024 Statewide Percentile Rank	2024 Lake Class Percentile Range	2024 Abundance Rating
87.8	75.2	15.4	12.9	47.8	76th	85th	High

RELATIVE ABUNDANCE (CPUE = NUMBER PER MILE) TRENDS

2013	2018	2024	Historical Average	2024 Statewide Percentile Rank	2024 Abundance Rating
22	35	9	22	49th	Moderate

AVERAGE AGE AT SPECIFIC LENGTH 8 INCH BIN

Species	Sex	Count	Average Length	Average Age	Age Range	Regional Rating	Lake Class
Yellow Perch	Male	1	8.1	7	7	Below Average	Below Average
Yellow Perch	Female	8	8.5	5.3	4 - 6	Average	Average
Yellow Perch	All	9	8.4	5.4	4 - 7	Average	Average

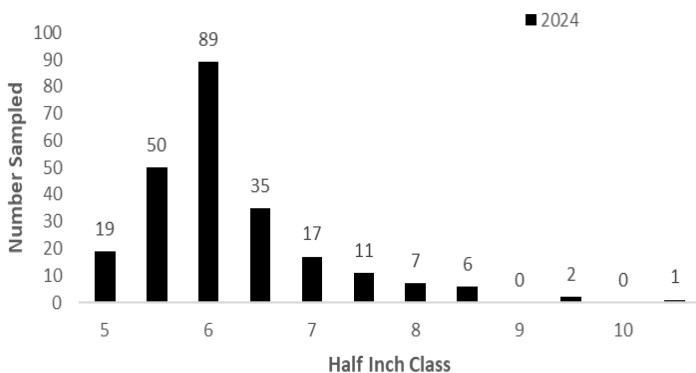


Photo Wisconsin DNR

Species Summary

- Little Green Lake in 2024 supports a moderately high population of yellow perch compared to statewide and similar lake class statistics, with catch rates of 12.9 per net night and 9 per mile of shoreline
- Size structure of yellow perch is moderately low based on data from the netting survey. Length data collected from the netting survey resulted in a PSD value of 7 which is in the 33rd percentile when compared to statewide values.
- The population is similar to the most recent survey in 2018 but down significantly from surveys in 2008 and 2013. Yellow perch seem to have an inverse relationship with black crappie in this lake and boom and bust opposite the black crappie population.
- Growth metrics calculated from age estimates indicate yellow perch have an average growth rate when compared across the state and as well as to lakes in the same complex warm dark lake class.

Yellow Perch Length Distribution - SN1 Little Green Lake





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Summary

Northern Pike

Northern pike numbers have gone down from our earlier surveys starting in 2008, but catch rates still fall in the 50th percentile compared to lakes in the same lake class. The average length of northern pike was 24.7 inches with males averaging 22.4 inches and females averaging 28.1 inches. The percentage of legal fish 26 inches and larger was 35%. The size structure and percentage of legal fish has almost tripled from 2008 when the population estimate was more than triple. So as the population has decreased the size structure has increased which is a sign that the population in 2008 was likely over abundant. With 2 fish in the 37 inch range and 16% 30 inches or larger, Little Green has the potential to grow some trophy northern pike. Growth rates were above average with fish reaching 26 inches after 5 years of growth. The 1.6 per acre population estimate is a little lower than the 2.0 per acre that Little Green could likely sustain. The slot northern pike regulation of no minimum size limit but fish between 25-35 inches must be released should be considered.

Walleye

Little Green Lake contains an above average walleye population at 2.3 adults/acre compared to lakes in the same lake class but it was still down from the last 3 surveys that averaged 3.3 adults/acre. Size structure was lower as well and likely affected by the large number of 10-12 inch fish in the population. The percentage of fish 20 inches and larger is 38% and near the surveys average of 41%. Walleye growth rates were above average with fish reaching the legal size of 15 inches during their 4th year of life. The oldest fish sampled was a 15+ year old 23.0 inch male. We floy tagged 188 walleyes 15 inches and larger with tag returns over the year giving a 13.4% exploitation rate. We were told that many anglers returned fish with tags back to the water without reporting them and we could see tags that had been scraped clean on fish we caught during our 2025 musky recap survey. Our work did not show significant exploitation and with the new 3 fish statewide bag limit change in 2024; no regulation changes are currently being proposed.

Muskellunge

A total of 66 muskies were caught in our survey. Catch rates have been relatively steady over the years ranging from 0.4-0.6 per net night. The population estimate of 0.4 fish per acre is in the 3rd quartile. Musky size structure was good with RSD40 up from recent surveys and similar to the 2003 survey. There have been no 50 inch fish sampled since 2003 however the average length is up from past surveys to 39.9 inches. This is likely due to the sample size of large females comprising nearly 50% of the catch as compared to 28% of the catch in 2003. No natural reproduction has been documented. Therefore stocking of 1 fish per acre is recommended to continue during odd years. Little Green Lake will continue to be managed as an intermediate density Class B musky fishery.

Largemouth Bass

Little Green Lake supports a moderately high density population of largemouth bass. At 32.2 per mile of shoreline it ranks in the 75th percentile when compared to lakes statewide in the same lake class. Size structure based off our SE2 survey ranked in the 80th percentile. This higher ranking is due to a low percentage of fish less than 12 inches in our sample. Fish sampled ranged from 6.5—19.4 inches and have above average growth rates when compared to same lake class averages. They are taking just over 5 years to reach the legal size of 14 inches.

Black Crappie

Black crappies, known for their boom and bust population swings, had an extremely high relative abundance from our netting survey when compared to previous surveys. Much of the population was made up of 4-5 inch fish and size structure was likely low and in the 8th percentile because of this large number of small fish. Growth rates was below average with it taking almost 8 years for a black crappie to reach 8 inches in length. During our netting survey we noticed many of the black crappie with patches of water mold (saprolegnia) with a small fish kill taking place as well. A sample of these fish was sent to our fish health experts and they confirmed the water mold along with some chronic inflammation in multiple organs with the cause of the inflammation not determined. Overabundance and poor condition of these fish likely contributed to their susceptibility. The average relative weight of the 8 and 9 inch fish was 79%, and an average healthy relative weight is 100%. Fish ranged from 4.2 - 9.8 inches and averaged 5.5 inches. Stocking of black crappie should be discouraged to let the population naturally stabilize. Management goal is to increase PSD8 $\geq$ 40, increase relative weight to around 100 and for growth a fish should reach 8 inches in 3-4 years.



Musky caught during our spring musky recap survey.



Fyke netting on Little Green



Fisheries Biologist Scott Bunde with a Little Green walleye



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Summary

Bluegill

There is an abundant population of bluegill in Little Green Lake ranking in the 97th percentile statewide and 96th percentile compared to lake class. Relative abundance is up 59% from the survey in 2018. Size structure is low ranking in the 12th percentile. Sizes of bluegill sampled ranged from 1.7 –7.0 inches with an average size of 5.2 inches. Growth rates for bluegill are average with it taking fish 5.4 years to reach 6+ inches in length. Exploitation rates based off the limited amount of fish we tagged was negligible. Management recommendation is to have CPE ≥ 3 inches = 200 - 300/mile, PSD from 40-60 and increase RSD8 to 5-10%.

Pumpkinseed

The pumpkinseed population in Little Green is high ranking in the 95th-96th percentile compared to statewide and lake class percentiles. Relative abundance of 61 per mile is up from 39 in 2018 but below the average of 81.3 per mile. Size structure was poor with no fish sampled larger than 6 inches and much lower than in 2018 when the 41% were 6 inches and larger.

Yellow Perch

Little Green Lake supports a moderately high population of yellow perch when compared to statewide and same lake class statistics. Our netting showed a similar population to the 2018 netting survey. Though down significantly from the 2008 and 2013 surveys. Size structure has been moderately low over the years and continues to be low with yellow perch falling in the 33rd percentile when compared to statewide values. Fish ranged from 5.1 -10.9 inches and averaged 6.4 inches. Growth rates are average when compared to lakes in the same lake class.

Other Species and Habitat

Other species sampled during our 2024 survey were brown bullheads, common carp, golden shiners, green sunfish, white suckers and yellow bullheads. Nearshore habitat such as weeds and especially wood are critical to bass survival and success, especially in lakes containing predatory fish such as muskellunge. The lake association is presently planning a pilot study using early top cutting of curly leaf pondweed. The goal is to open up the plant canopy for native vegetation without significantly impacting young fish. Fisheries management will be monitoring the project closely.

Stocking History

Stocking Year	Species	Number Fish Stocked	Avg Fish Length(IN)	Stocking Year	Species	Number Fish Stocked	Avg Fish Length(IN)
2024	WALLEYE	2000	7	1999	MUSKELLUNGE	876	10.9
2024	YELLOW PERCH	5000	4	1998	WALLEYE	10000	1.7
2024	FATHEAD MINNOW	800	2	1997	WALLEYE	20200	2.7
2023	WALLEYE	14264	2.55	1996	MUSKELLUNGE	932	11.8
2021	YELLOW PERCH	6000	5	1995	WALLEYE	1026703	1.65
2021	NORTHERN PIKE	100	14	1993	MUSKELLUNGE	1000	9
2021	MUSKELLUNGE	462	10	1992	MUSKELLUNGE	930	10
2021	WALLEYE	1500	7	1991	MUSKELLUNGE	1200	11
2020	WALLEYE	2683	7.45	1991	WALLEYE	12060	3
2020	WALLEYE	8875	1.69	1990	MUSKELLUNGE	1200	10
2019	MUSKELLUNGE	932	11.58	1990	WALLEYE	35680	4
2019	WALLEYE	17797	2.05	1989	MUSKELLUNGE	400	7
2018	WALLEYE	2600	7	1989	WALLEYE	23584	2
2018	MUSKELLUNGE	466	12	1988	NORTHERN PIKE X MUSKE	1600	7.5
2017	WALLEYE	16174	1.7	1988	WALLEYE	3535	3
2017	WALLEYE	3682	8	1988	MUSKELLUNGE	400	11
2017	MUSKELLUNGE	500	6.3	1987	WALLEYE	72000	2
2017	MUSKELLUNGE	932	9.3	1987	NORTHERN PIKE X MUSKE	2400	7
2016	MUSKELLUNGE	466	10.3	1987	MUSKELLUNGE	1200	3
2016	FATHEAD MINNOW	10000	1	1986	MUSKELLUNGE	400	8
2016	BLACK CRAPPIE	1000	4	1986	NORTHERN PIKE X MUSKE	800	9
2016	WALLEYE	1995	6.5	1985	NORTHERN PIKE X MUSKE	800	10
2016	NORTHERN PIKE	168	9	1985	MUSKELLUNGE	400	8
2015	WALLEYE	16184	1.8	1984	MUSKELLUNGE	400	8
2014	MUSKELLUNGE	466	9.5	1984	NORTHERN PIKE X MUSKE	800	8
2013	MUSKELLUNGE	466	11.35	1984	WALLEYE	24050	3
2012	WALLEYE	23285	2	1983	NORTHERN PIKE X MUSKE	800	9
2012	MUSKELLUNGE	950	9.6	1983	MUSKELLUNGE	350	9
2011	WALLEYE	16741	2.1	1982	NORTHERN PIKE X MUSKE	800	9
2011	WALLEYE	1500	7	1982	MUSKELLUNGE	400	9
2011	FATHEAD MINNOW	150000	1	1981	MUSKELLUNGE	400	9
2011	MUSKELLUNGE	930	9.3	1981	NORTHERN PIKE X MUSKE	800	9
2011	MUSKELLUNGE	100	12	1981	WALLEYE	24000	2
2010	MUSKELLUNGE	821	12.8	1980	NORTHERN PIKE X MUSKE	1600	7.5
2009	WALLEYE	15310	1.8	1980	MUSKELLUNGE	400	11
2009	WALLEYE	3664	6	1979	MUSKELLUNGE	436	9
2009	MUSKELLUNGE	931	10.2	1979	NORTHERN PIKE X MUSKE	800	10
2008	MUSKELLUNGE	932	10.3	1978	MUSKELLUNGE	400	10
2008	MUSKELLUNGE	100	8	1978	NORTHERN PIKE X MUSKE	800	9
2008	WALLEYE	2000	5	1977	MUSKELLUNGE	1040	11
2007	MUSKELLUNGE	620	13	1977	NORTHERN PIKE X MUSKE	800	11
2007	WALLEYE	16415	2.1	1976	MUSKELLUNGE	400	13
2006	MUSKELLUNGE	205	10.8	1976	NORTHERN PIKE X MUSKE	826	6
2005	WALLEYE	24380	1.91	1975	NORTHERN PIKE X MUSKE	800	13
2005	MUSKELLUNGE	932	10.6	1975	MUSKELLUNGE	200	9
2004	MUSKELLUNGE	999	10.5	1974	NORTHERN PIKE X MUSKE	828	11
2003	MUSKELLUNGE	932	10.9	1974	MUSKELLUNGE	400	11
2003	WALLEYE	27032	2.5	1973	MUSKELLUNGE	1300	14
2002	MUSKELLUNGE	931	10.1	1973	WALLEYE	500	9
2002	NORTHERN PIKE X MUSKE	1000	10	1973	WALLEYE	1500000	
2001	NORTHERN PIKE X MUSKE	500	8.1	1972	MUSKELLUNGE	18000	1
2001	MUSKELLUNGE	932	10.6	1972	WALLEYE	500	9
2001	WALLEYE	23300	1.6	1972	MUSKELLUNGE	1200	15
2000	NORTHERN PIKE X MUSKE	1000	8	1972	WALLEYE	1000000	1
1999	WALLEYE	25000	1.5	1972	WALLEYE	2638	7
1999	NORTHERN PIKE X MUSKE	200	8.1				